



Figure: X11 forwarding using SSH

system's *messages* file. To access it, you must log in as the *root*, so the command to use is as shown below:

```
ssh root@192.168.1.100 'tail -f /var/log/messages'
```

Now what's been done so far is text-UI-based; can we use the GUI over SSH? Yes, by using a technique called X11 session forwarding.

X11 session forwarding

This is designed to transmit window and bitmap information over the network, and lets you log in to a remote machine and run an XWindow application (like GIMP, OpenOffice, games, etc). While the application runs on the remote computer, the GUI will be displayed on your local computer. The `-X -f` switches to SSH enable this; you run something like `ssh -X -f username@host application_name` to accomplish this. `XClock` is a good application to test this; run `ssh -X -f binay@192.168.1.100 xclock` and enter the password. You should then see the clock window on your screen, as seen in the Figure. If this does not work, please check in `/etc/ssh/sshd` whether X11 forwarding has been set to 'Yes'. After verifying that X11 forwarding works, you can run any GUI application remotely—for example, the Nautilus file manager, or even a game like Mahjong. You can even run a whole desktop environment (DE) on the remote machine, viewed on your local machine, with a command like `ssh -X -f binay@192.168.1.100 gnome-session`. You should see the GNOME panel appear on your local display.

Finally, you can end your SSH session by pressing `Ctrl-D`.

This is just a brief introduction to SSH. It is not limited to LAN use only; you can use SSH to connect to computers across the globe. There is more interesting information on SSH tunnelling and port forwarding, which I will cover in an upcoming issue. 

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